

Name:Sudiptika J Register Number: 250701754 Experiment Name:K-Means Clustering

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

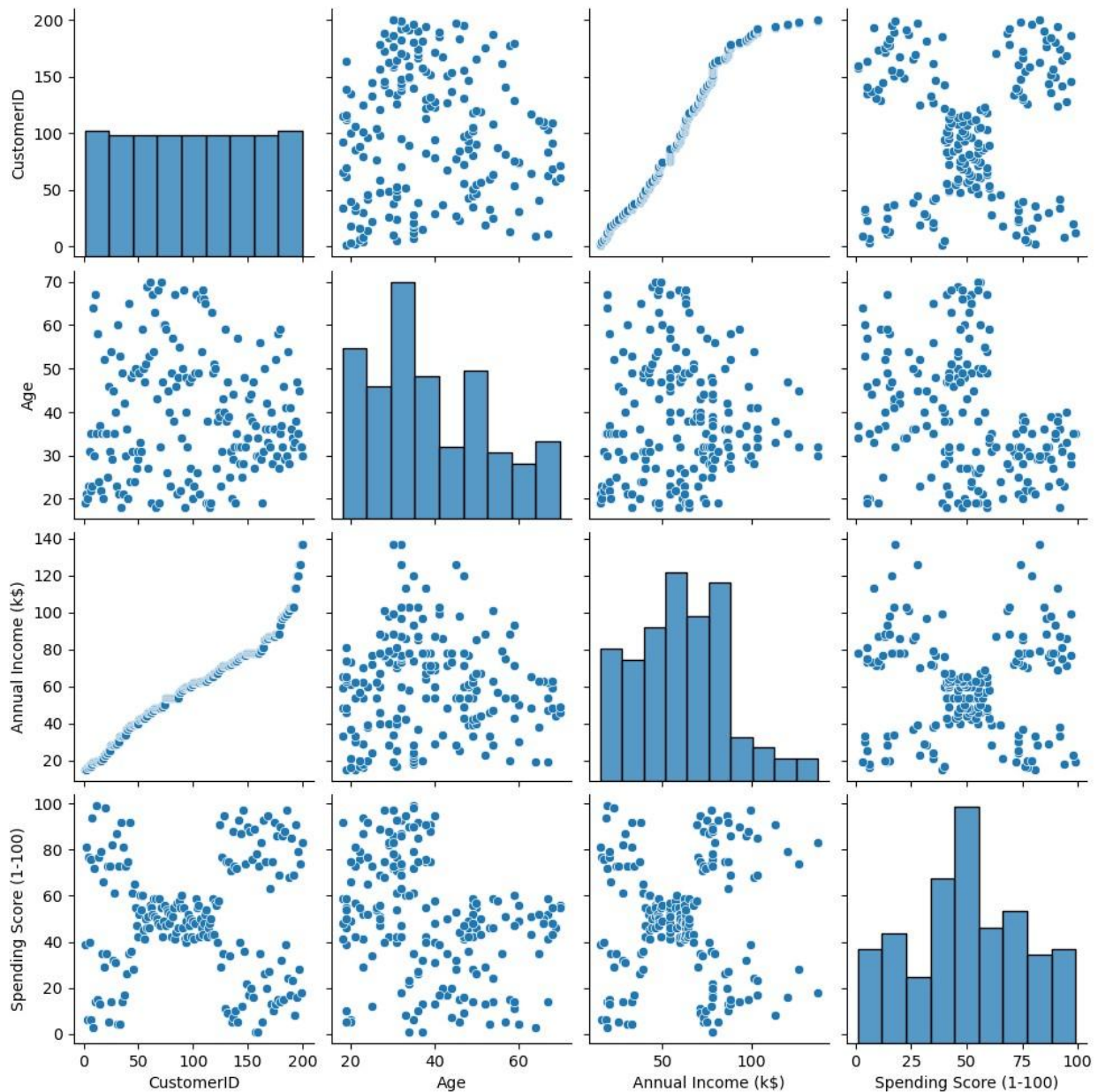
df=pd.read_csv('Mall_Customers.csv')

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                          200 non-null    int64
1   Gender                              200 non-null    object
2   Age                                 200 non-null    int64
3   Annual Income (k$)                  200 non-null    int64
4   Spending Score (1-100)              200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
df.head()

   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1   Male   19                15                39
1           2   Male   21                15                81
2           3  Female   20                16                 6
3           4  Female   23                16               77
4           5  Female   31                17               40
sns.pairplot(df)

<seaborn.axisgrid.PairGrid at 0x1f67ce53230>
```



```
features=df.iloc[:,[3,4]].values
```

```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

C:\Users\swath\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable

```

OMP_NUM_THREADS=1.
warnings.warn(
KMeans(n_clusters=5)

Final=df.iloc[:, [3,4]]
Final['label']=model.predict(features)
Final.head()

C:\Users\swath\AppData\Local\Temp\ipykernel_5828\470183701.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#
returning-a-view-versus-a-copy
Final['label']=model.predict(features)

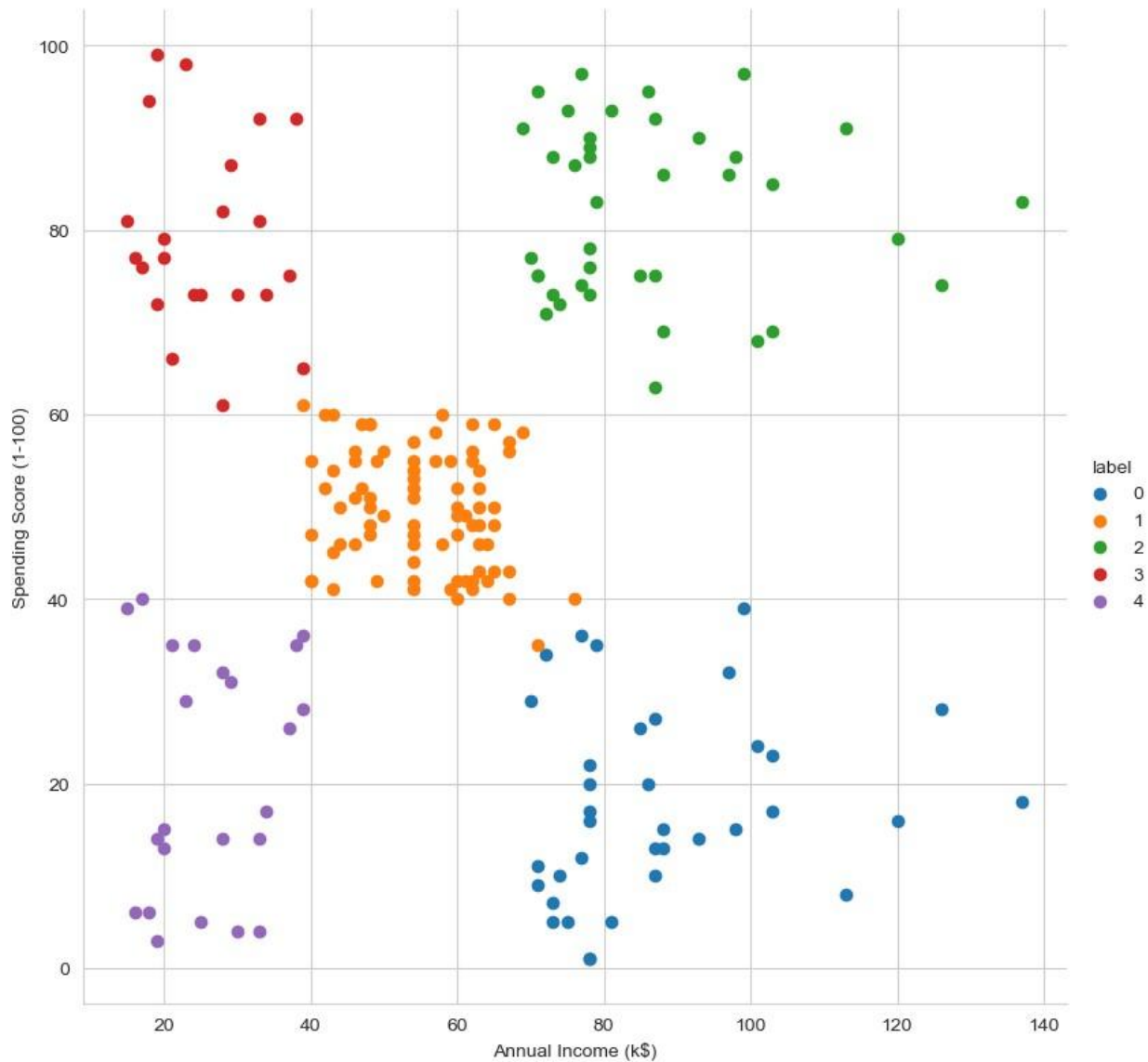
```

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	3
2	16	6	4
3	16	77	3
	17	40	4

```

sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()

```



```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Cell In[11], line 5

```
model=KMeans(n_clusters=i)
^
```

IndentationError: expected an indented block after 'for' statement on line 4

